SUPERMART, INC.

Customer Segmentation and Marketing Campaign Analysis

Understanding spending patterns to improve marketing strategy

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SUPERMART, INC.

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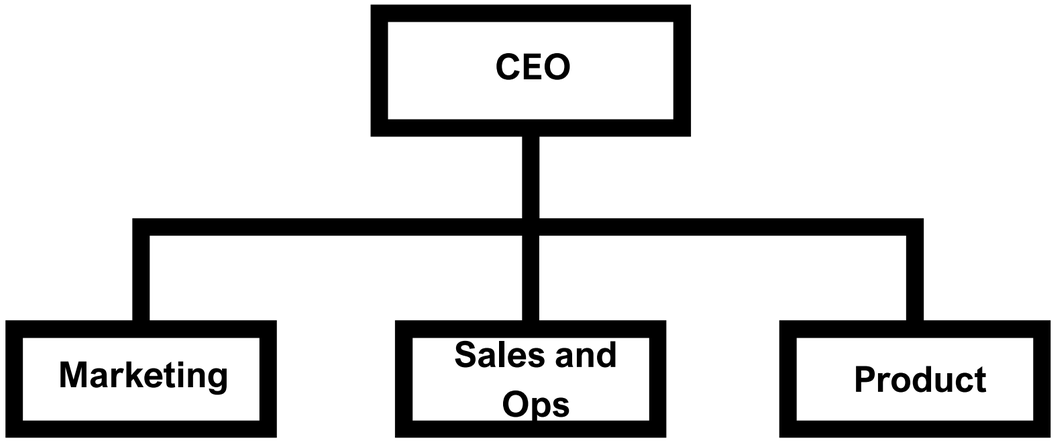


Background: Business Context

Supermart, Inc. is a retail supermarket company that provides a wide range of daily consumer products, including fresh food, household essentials, and premium goods such as wine and meat.

Supermart exists to fulfill customers' daily needs by offering convenient access to essential goods, while also providing variety, quality, and competitive pricing.

Background: Business Stakeholder



Background: Understanding Business

Supermarket collect large amounts of customer data but often lack insight into how different customer segments behave.

Through data analysis and visualization, the project aims to:

- ***Uncover patterns in customer behavior***
- ***Identify opportunities for improving marketing strategies and product focus***

Business Statement

How can customer segmentation based on demographics, shopping behavior, and campaign responses be used to develop more targeted and effective marketing strategies?

The insights from this analysis will help the supermarket optimize campaign targeting, prioritize high-value segments, and improve overall marketing performance.



Data Overview

The dataset used in this project contains information about supermarket customers, including their demographic characteristics, purchasing activity, and responses to marketing campaigns. Each row represents an individual customer, while the columns describe various aspects of the customer's profile and behavior.

Customer Demographic

- Year_Birth – customer’s birth year
- Education – highest education level
- Marital_Status – marital status
- Income – yearly household income
- Kidhome – number of children in the household
- Teenhome – number of teenagers in the household
- Recency: Number of days since customer's last purchase

Marketing Campaign Response

- AcceptedCmp1 to AcceptedCmp5
- Response (response to the most recent campaign)

Customer Spending Behavior

- MntWines
- MntFruits
- MntMeatProducts
- MntFishProducts
- MntSweetProducts
- MntGoldProds

Purchase Channels

- NumWebPurchases
- NumCatalogPurchases
- NumStorePurchases
- NumWebVisitsMonth

Data is taken from Supermarket Customers.csv, shared in March 2026, with customer enrollment dates ranging from July 2012 to June 2014.

Data Understanding

2,240

Total Customer

~$52,000

Average Income

~$603

Average Spending

7 %

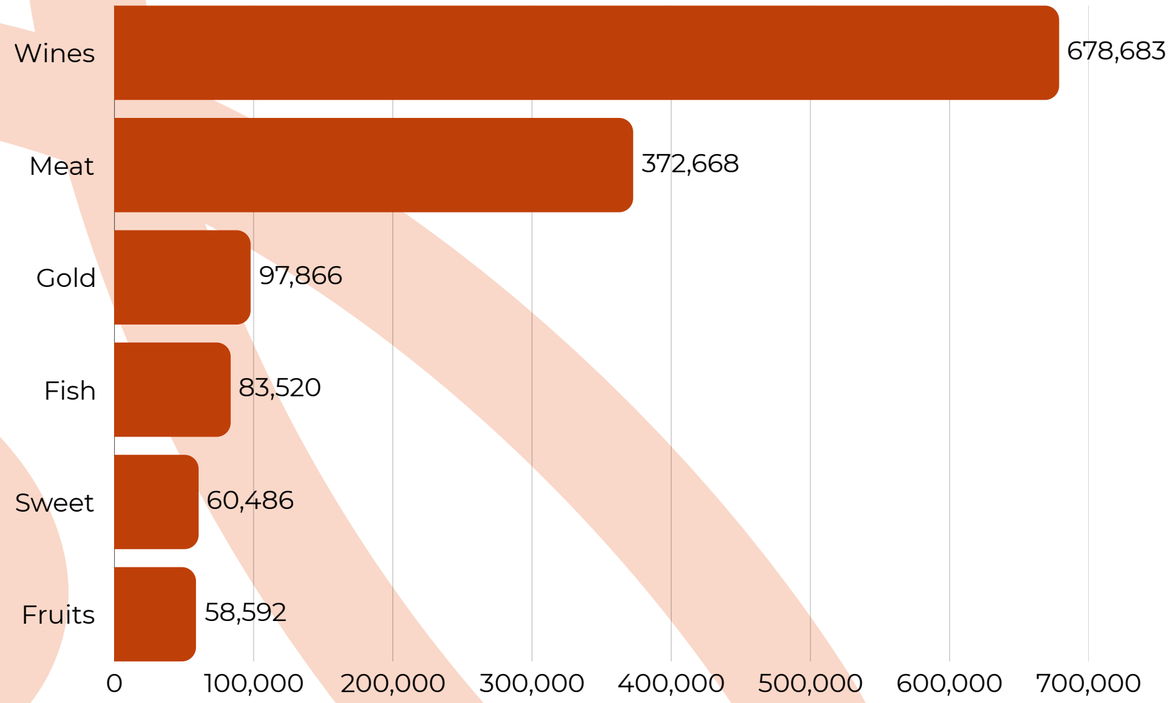
Overall Campaign Response

Exploratory Data Analysis (EDA)



Exploratory Data Analysis (EDA)

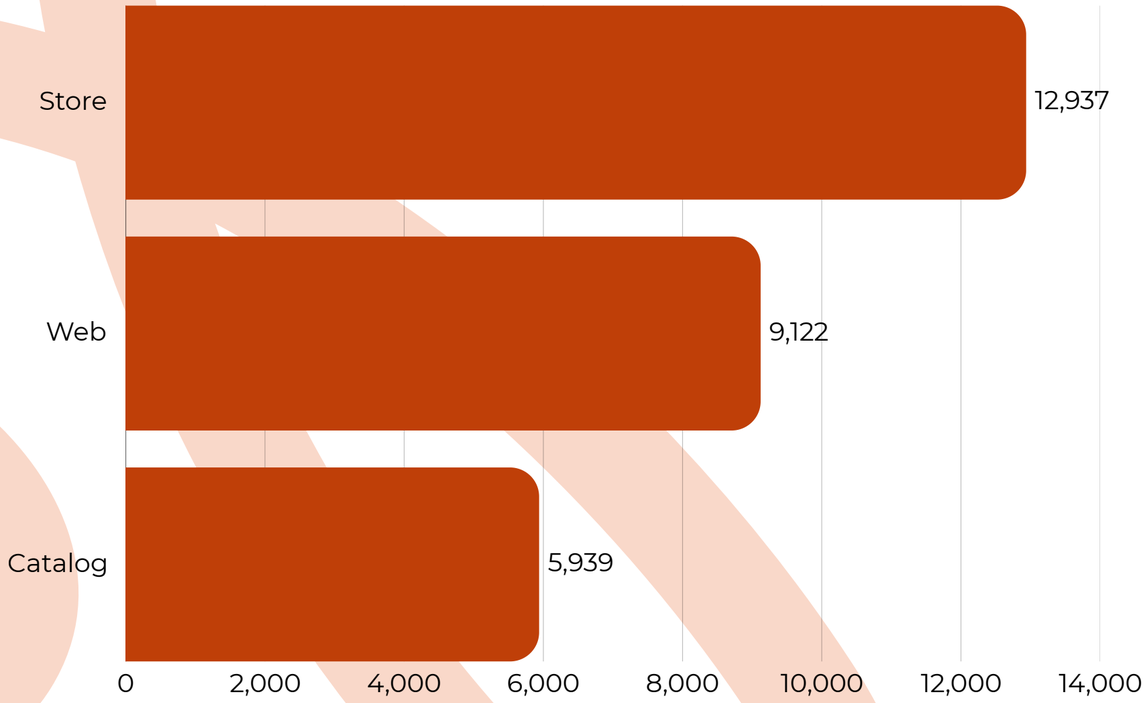
Customer Spending by Product Category



- **From the visualization, it can be observed that wine and meat products contribute the largest portion of customer spending, while fruits and sweet products contribute significantly less.**
- **This indicates that these categories may represent high-value product segments for the supermarket.**

Exploratory Data Analysis (EDA)

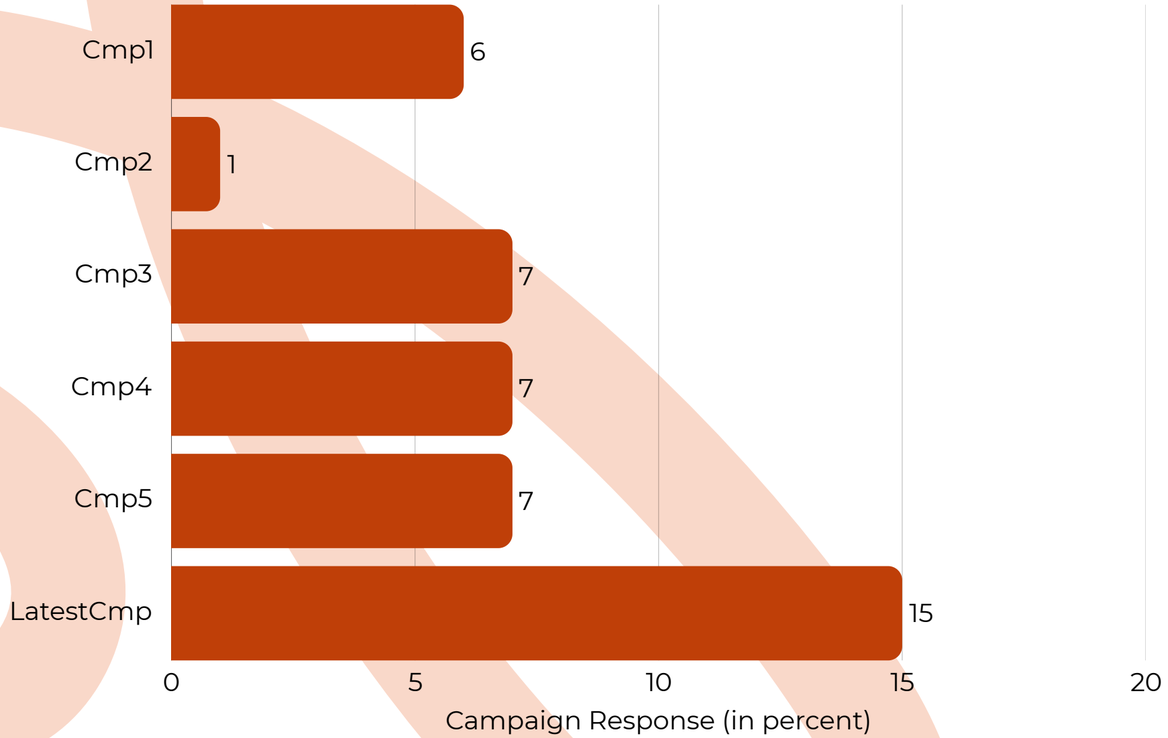
Purchase Channel Preference



- **The visualization shows that store purchases represent the largest share of transactions, followed by web purchases and catalog purchases.**
- **This suggests that physical stores remain the primary sales channel, although digital channels still play an important supporting role.**

Exploratory Data Analysis (EDA)

Marketing Campaign Acceptance



- **The acceptance rates for most campaigns appear relatively low (1-7%). This suggests that many marketing campaigns may not be effectively targeting the right customers.**
- **However, the latest campaign shows a noticeably higher acceptance rate (~15%) compared to previous campaigns, which may indicate improvements in campaign targeting or marketing strategy.**

Diagnostic Analysis



Diagnostic Analysis

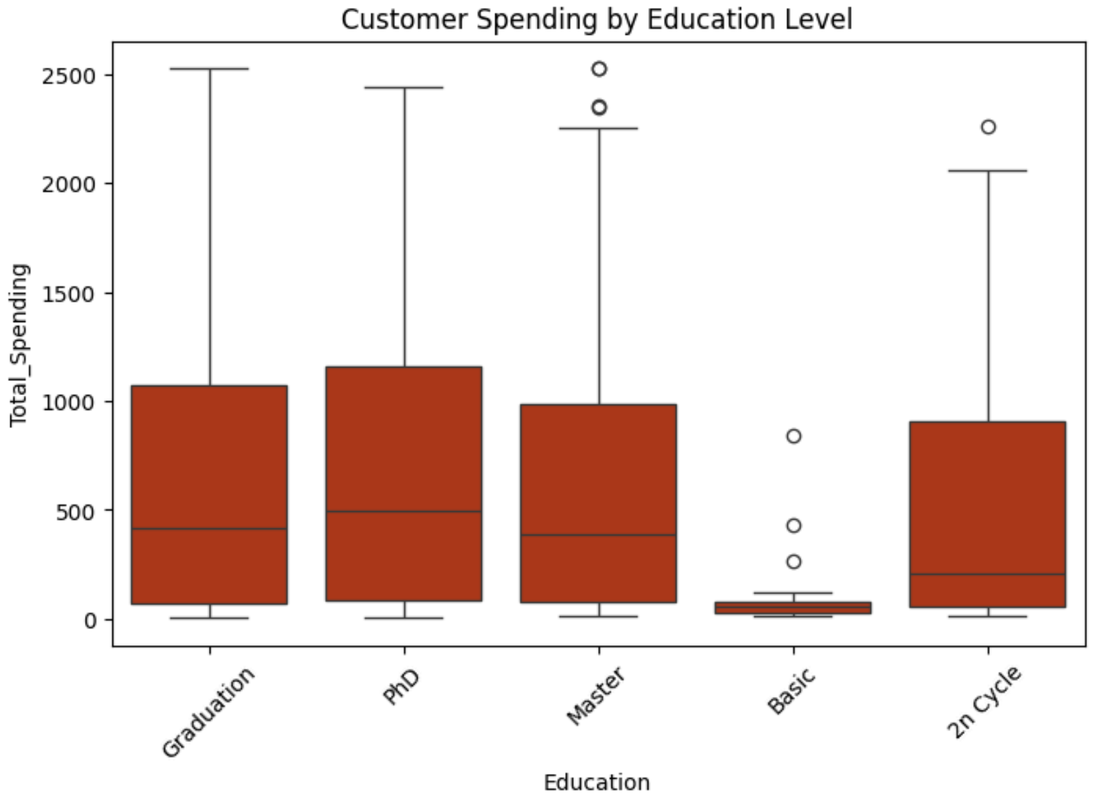
Income vs Customer Spending



- The scatter plot shows a positive relationship between income and total spending, with correlation test showing strong relationship.
- Customers with higher income levels tend to spend more across product categories. This suggests that income is an important factor in customer segmentation.

Diagnostic Analysis

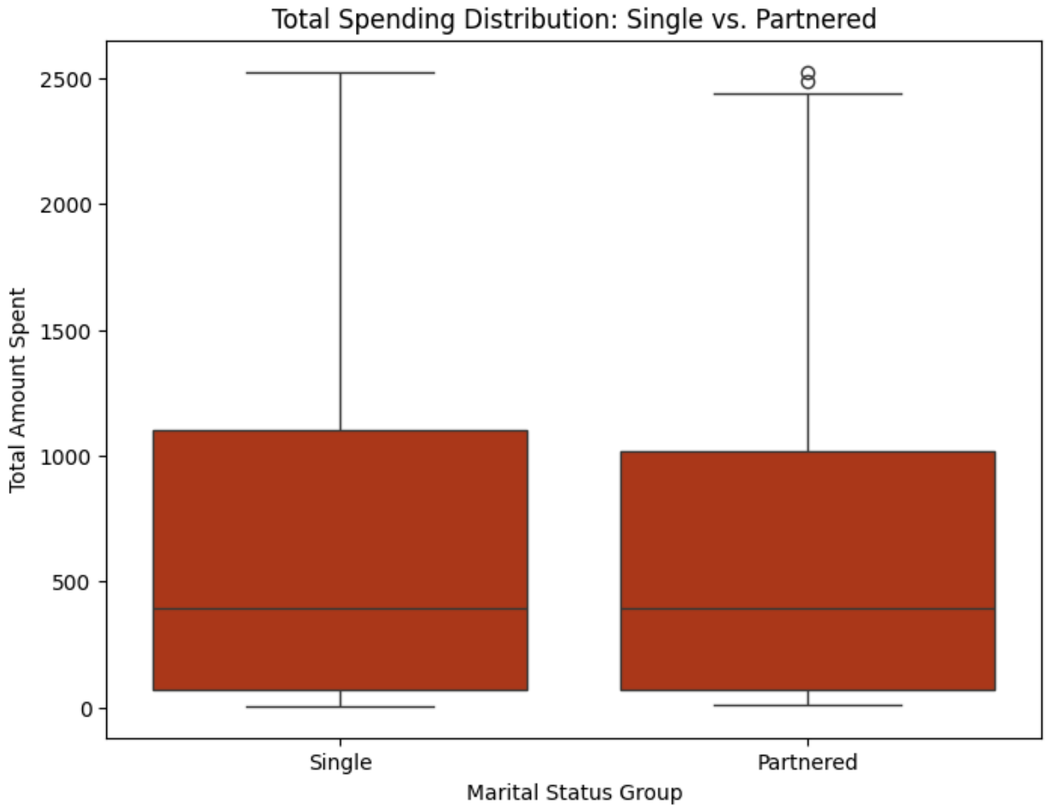
Spending by Education Level



- The distribution shows that customers with higher education levels tend to have higher median spending.
- This suggests that education may have some influence on spending behavior.
- Most customers have higher education levels, while customers with basic education are only as small as 2.42%, which is negligible.

Diagnostic Analysis

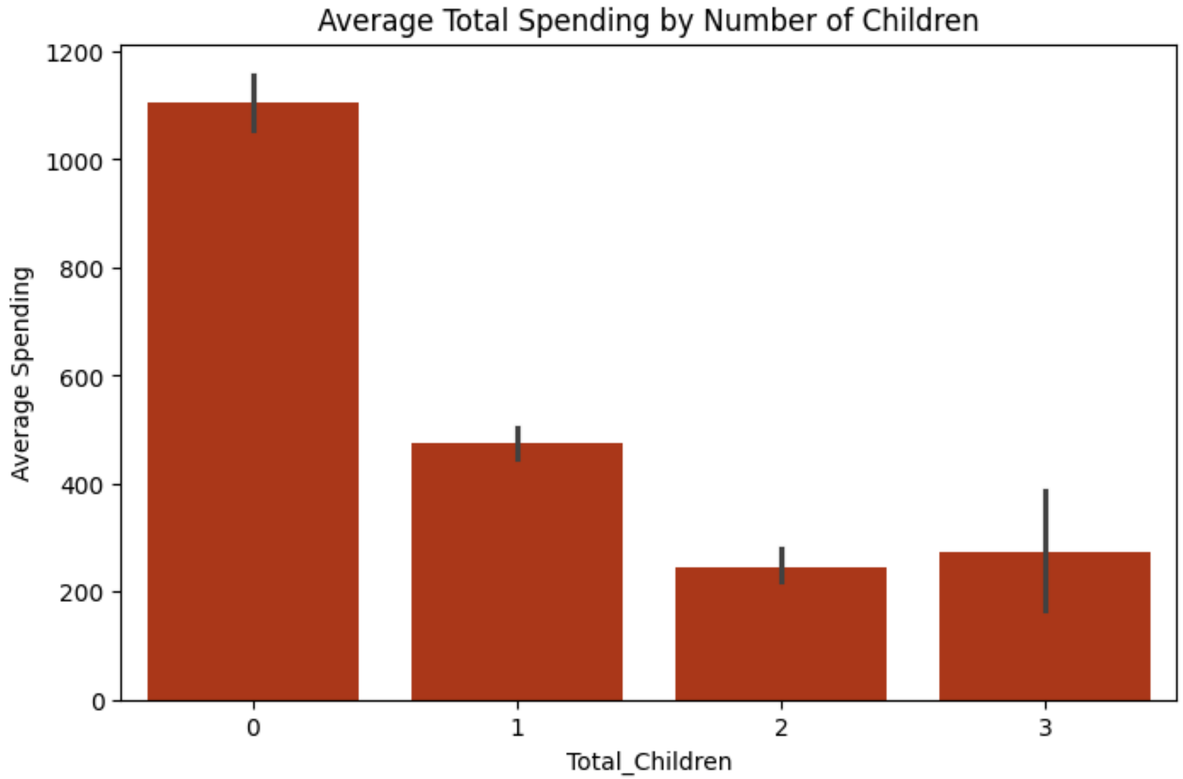
Spending by Marital Status



- **Marital status (Single vs. Partnered) shows a negligible impact on total spending.**

Diagnostic Analysis

Spending based on Children in Families



- **While marital status (Single vs. Partnered) has a negligible impact on spending, household size (children) is a primary driver of purchasing behavior.**
- **Customers with no children spend significantly more on our product categories than those with 1 or more children.**

Diagnostic Analysis

Customer Segmentation

	Income_Group	Spending_Group	Has_Children	Total_Spending	Response	Customer_Count
5	Low	High	Yes	1730.000000	0.000000	1
16	High	High	No	1484.117359	0.330073	409
17	High	High	Yes	1256.700980	0.127451	204
10	Medium	High	No	1207.875000	0.312500	32
11	Medium	High	Yes	1055.666667	0.135417	96
4	Low	High	No	837.000000	0.500000	2
14	High	Medium	No	690.206897	0.206897	29
15	High	Medium	Yes	587.427083	0.052083	96
9	Medium	Medium	Yes	413.547847	0.105263	418
8	Medium	Medium	No	403.127660	0.106383	47
2	Low	Medium	No	295.500000	0.131579	38
3	Low	Medium	Yes	238.008696	0.226087	115
6	Medium	Low	No	100.000000	0.000000	1
13	High	Low	Yes	84.000000	0.000000	4
7	Medium	Low	Yes	56.126667	0.013333	150
0	Low	Low	No	50.306667	0.080000	75
1	Low	Low	Yes	49.772374	0.093385	514
12	High	Low	No	7.000000	0.000000	2

- This segmentation focuses on three primary variables: income, total spending, and household composition (presence of children), as these factors have been identified as key drivers of differences in customer behavior.
- High-spending segments consistently appear at the top, and are generally associated with higher income levels, followed by medium- and low-income groups.
- In contrast, the presence of children is more evenly distributed across segments and does not follow a clear pattern.
- Customer count is a critical consideration when evaluating segments. Segments with very small populations should be excluded from targeted strategies, as they are likely to have limited impact on overall business performance.

Diagnostic Analysis

Customer Segmentation

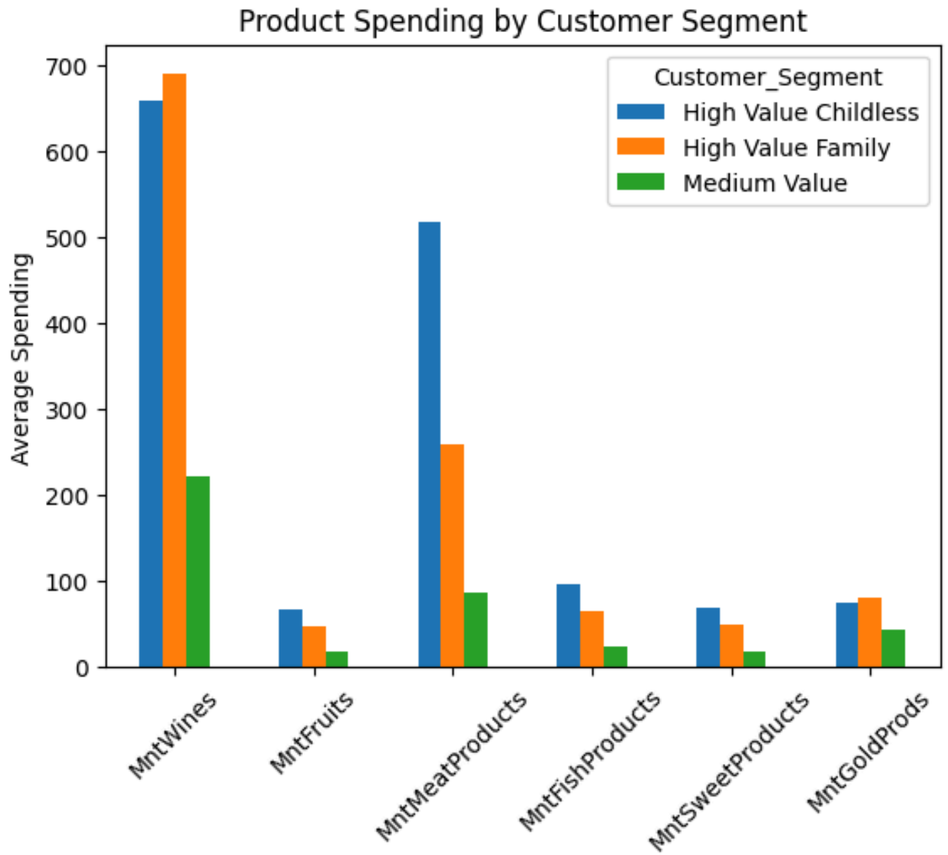
Customer_Segment	Total_Spending	Response
High Value Childless	1484.117359	0.330073
High Value Family	1193.334328	0.149254
Medium Value	412.946164	0.122476

- With this cleaner segment summary, it is easier to understand the case. It is best to focus on few segments rather than lots of segments, so I use three segments in here. They are:

- High Value Childless. The high-income, high-spending customers without children represent the most valuable segment.
- High Value Family. High-income customers with children also demonstrate strong spending behavior.
- Medium Value Customers in the medium spending segment, regardless of income level, show moderate purchasing behavior.

Diagnostic Analysis

Segmentation Validation



- **Wine & Meat = Top Revenue Drivers**

→ Highest spending across all segments
→ Explore bundling opportunities

- **High-Value Childless Segment → Premium Opportunity**

→ Offer smaller, premium portions
→ Higher margins, less “waste concern”

- **Focus on Core Strengths**

→ Double down on wine & meat categories
→ Optimize via promotions & product variety



Conclusion

Conclusion

Key Insights	Operational Implication	Revenue & Customer Impact
Customer Spending is Driven by Income & Household Composition	• High-income customers contribute most revenue • One-size-fits-all marketing is ineffective	• Risk of inefficient targeting • Missed opportunities to maximize high-value segments
High-Value Customers Are Not Homogeneous	• Childless customers are highly responsive • Families spend high but respond less	• Lost revenue from under-targeted segments • Lower campaign effectiveness
Spending Does Not Equal Campaign Engagement	• High spenders are not always responsive	• Wasted marketing budget • Low campaign ROI if not segmented properly
Wine & Meat Dominate Customer Spending	• Core revenue driven by specific categories	• Opportunity to increase basket size and margins



Recommendation

Recommendation

Customer Segment / Insight	Marketing Action Plan	Revenue & Customer Impact
Supermart, Inc. should adopt a customer segmentation approach based on demographics, purchasing behavior, and campaign responsiveness. By replacing one-size-fits-all campaigns with segment-specific strategies, the company can improve targeting accuracy, increase engagement, and maximize marketing ROI.		
High-Value, High-Response (Childless, High Income)	• Loyalty programs & exclusive offers • Premium bundling (wine & meat, smaller portions) • Personalized campaigns	• Increase retention & repeat purchase • Higher profit margin from premium pricing
High-Value, Low-Response (Families with Children)	• Family-oriented bundles (larger size) • Adjust messaging to family needs	• Unlock untapped revenue potential • Increase engagement in valuable segment
Medium-Spending, Low Engagement (Growth Segment)	• Targeted discounts & promotions • Upselling strategies	• Increase spending per customer • Convert into high-value customers • Drive overall revenue growth

Thank You

Appendix - Data Cleaning

Issue Identified	Action Taken	Impact on Analysis
Missing values in Income	Removed or imputed missing values	Ensures accurate segmentation and avoids bias
Date format in Dt_Customer	Converted to datetime format	Enables time-based understanding of customer data
Checked duplicates	Removed duplicate records (none were found)	Ensures data integrity
Outliers in income	Reviewed and kept (justified)	Prevents distortion in spending and segmentation analysis
Irrelevant variables (ID, Z_CostContact, Z_Revenue)	Removed columns with constant or non-informative values	Reduces noise and focuses analysis on meaningful features
Invalid categories in Marital_Status (Absurd, YOLO)	Removed rows with unrealistic values	Improves data quality and reliability

Appendix - Data Transformation

Transformation	Method	Purpose / Impact
Total Spending	Aggregated all product spending variables (Mnt*)	Creates a single metric to measure overall customer value
Customer Segmentation	Grouped customers into Low, Medium, High (income & spending) + Children flag	Enables targeted marketing analysis
Income Grouping	Categorized income into Low, Medium, High	Simplifies comparison across customer groups
Spending Grouping	Categorized total spending into tiers	Identifies high-value vs low-value customers
Children Flag	Combined Kidhome & Teenhome into binary variable (Has Children / No Children)	Captures household composition more effectively
Campaign Response Indicator	Aggregated campaign responses	Measures overall engagement level